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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

17 October 2026

Hiring Team
Scarlet

Application: Machine Learning Engineer (Applied Machine Learning)

Dear Scarlet hiring team,

I am applying for the Machine Learning Engineer role in Applied Machine Learning. For three years at HDFC Asset Management I took GenAI, retrieval and agentic systems from prototype to production for Compliance, Client Service and Investment teams, working directly with those domain experts to decide what to build and changing each system on what they found in use. Helping assessors find and check evidence in large volumes of messy documents without lowering the safety bar is close to the work I did with a Compliance department in a regulated financial firm. I do not know medical device certification and would expect to learn it from your assessors.

At HDFC I designed and built the applications end to end: a ReactJS frontend, Python FastAPI microservices and RAG with LlamaIndex, Azure OpenAI embeddings and a separate pgvector database per department, so that each team retrieved only from its own documents, with role-based access through the company LDAP server. Agents called Python functions as tools to run SQL-backed fund comparisons and generate PDF reports. Compliance reviewed every release, so I guardrailed behaviour with system prompts and versioned every prompt change in Git. On security I agreed port whitelisting with the security team, restricted access to private IPs with AWS Security Groups and deployed an Nginx reverse proxy, one of the fixes that took the applications from repeatedly failing to passing their VAPT. I also ran a pipeline that parses over 300 rating-agency emails a day, first with AWS Textract and then with Azure OpenAI when one agency began sending emails as images, saving analysts 60 hours a month.

In my AstraZeneca MSc dissertation I co-built CellLineSelector, which ranks cancer cell lines and returns the evidence behind each pick; my focus was the data preprocessing, the scoring and ranking algorithm and the RAG layer, and we refined and validated the ranking with feedback from AstraZeneca scientists. In the VisionCaptioner group project we compared encoder and decoder architectures across five seeds, scored with SBERT cosine similarity, BLEU and ROUGE-L.

On your list, I have not built a formal evaluation framework: at HDFC quality came from user feedback, system-prompt guardrails and Compliance review rather than golden datasets or LLM-as-judge, and I have not measured citation correctness or agreement with expert judgment. I have not built custom agent harnesses for very large document collections, and I have about three years of experience. Evals and agentic retrieval over large document sets are where I would expect to grow fastest.

What I can offer is the habit of owning an ambiguous problem with domain experts. The first GenAI systems at HDFC were a two-person effort, my manager and me, and I worked directly with Client Service, Compliance and Investment to define what to build, release it and improve it; in my final year I acted as de facto team lead for up to four interns. Python is my day-to-day language.

I am currently based in Bristol, hold a Student Visa and am eligible for the Graduate Visa, which gives two years of full working rights. The role is based in your London office and I would be glad to discuss working from there. Thank you for considering my application; I would be glad to talk it through.

Yours sincerely,
Kaustubh Sonawane